Assignment 2: Detecting emotions with ML algorithms

Introduction

Expressing emotions is one of the things that make us humans. To recognize emotions from facial expressions is a task which in computer vision is often referred to as “facial expression recognition” or “emotion recognition”. In this project you will be performing emotion recognition by training machine learning algorithms that operate on images.

The goal of this assignment is to contrast the challenges associated with the datasets and models for the same task. Specifically, given two datasets used for emotion recognition. Which unique challenges are needed to be addressed to build a successful emotion recognition with each of them? Can you leverage any data, insights, features that you can leverage from one dataset to improve the performance of models evaluated on another one?

Goal

The goal of this project is to understand the challenges of generalization across datasets. This assignment considers the use of two datasets for emotion recognition. The first dataset is the FER 2013 dataset, which contains still images with images displaying different 7 emotions. KDEF dataset, which also contains facial images displaying different expressions. Although both datasets were designed with similar objectives in mind, you will be able to observe that the performance of models trained on each of these models differ considerably. Moreover, you will have to train models on each of these datasets and evaluate on the other one. In addition, you develop techniques to extract information from the KDEF dataset to help you improve the performance in the FER 2013 dataset.

Note that the FER→KDEF and KDEF→FER transfer scenarios constitute a significant domain adaptation problem. The two datasets differ with respect to (i) face cropping and alignment procedures, (ii) image resolution, and (iii) color representation (grayscale versus RGB). These factors induce distribution shifts in the input space that can negatively affect generalization performance independently of the emotion recognition task itself. Therefore, cross-dataset results should be interpreted as a measure of both emotion recognition capability and robustness to dataset-specific acquisition and preprocessing differences.

Datasets

FER 2013 dataset [R1]

The FER 2013 dataset has the following characteristics

- The images are 48x48 grayscale patches which contain faces expressing the emotions labeled as “angry”, “disgust”, “fear”, “happy”, “sad”, “Surprise”, “neutral”.
- There are different versions of this dataset in kaggle. Use the version where all the dataset is stored as a singles .csv file, in which the numerical values of the images are stored, as well as the class labels (https://www.kaggle.com/datasets/deadskull7/fer2013). This version is easier to use,

as you can read it out of the box with pandas's read_csv routine. You may need to reshape the image data to 48x48 images for feature extraction.

- Use the FER2013 training split for model training. Use the PublicTest (validation) split for model selection and hyperparameter tuning. The PrivateTest (test) split must only be used for the final evaluation and reporting of the selected models. In the given data splitting notebook, the referred data splits will be referred to as training, validation and test.
- Report the performance on the private test set.
- We provide with a script to read and split this dataset.

KDEF Dataset (Karolinska Directed Emotional Faces) [R2]

The KDEF (Karolinska Directed Emotional Faces) dataset is a facial expression dataset created by researchers at the Department of Clinical Neuroscience, Karolinska Institutet. It contains photographs of human faces displaying a set of basic emotions under controlled acquisition conditions. The dataset can be download here: https://www.kdef.se/download-2/register.

Main characteristics

- 140 subjects (70 female and 70 male).
- 7 facial expressions:
- Afraid
- Angry
- Disgust
- Happy
- Neutral
- Sad
- Surprise

5 viewing angles (poses):

- Full Left (FL)
- Half Left (HL)
- Straight/Frontal (S)
- Half Right (HR)
- Full Right (FR)
- **For this assignment you are only required to use the straight/frontal pose (S)**

Approximately 4,900 images in total (7 emotions × 5 poses × 140 subjects)

Tasks

1. **FER2013 Baseline (20%). As a starting point you will be given a Random Forest + Landmark feature extraction.**
    a. Build a baseline FER system.
    b. Compare at least:
        i. Logistic Regression

ii. SVM
iii. Random Forest

c. Compare at least the following feature types
   i. HOG
   ii. Facial landmarks
   iii. PCA features

d. Report
   i. Accuracy, balanced accuracy
   ii. Macro-F1
   iii. Confusion matrix

For your report

- Explain which classifier performs best
- Which features work best
- Explain which emotions are most difficult to classify
- Provide some evidence supporting your arguments when needed (e.g. examples of samples difficult to classify, images with variation in light, mislabelled examples, etc.)

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2. KDEF Dataset Study (15%).

Use the best feature representation identified on FER2013 and retrain the corresponding classifiers on KDEF.

- The KDEF contains multiple images of the same individual, the evaluation must be subject-independent.
- Use an 80/20 split based on subject identity, such that all images from a given subject belong either to the training set or to the testing set, but never both.
- Remember to use only the frontal images.
- Use included pre-processing script for adapting images of KDEF to a format closer to the FER 2013 dataset.

For your report

- Why are KDEF images easier/harder to classify than FER2013?
- Which models change ranking?
- Which emotions transfer well between datasets?

3. Cross-Dataset Generalization (20%)

Select the best-performing configuration from Task 2. Retrain this configuration using the preprocessed KDEF images and evaluate it directly on FER2013.Questions to answer in your report

- What adaptations did you have to make to run the model?
- What is domain shift?
- Why does performance degrade?
- Which features transfer best?

4. Explainable Fuzzy Classifier (10%)

- Build Fuzzy classifier on the KDEF dataset
- Use a limited set of 3-10 handcrafted geometric features. For example
    - mouth opening
    - mouth width
    - eye opening
    - eyebrow height
    - smile ratio
- Landmarks may be extracted using OpenCV or dlib.

For your report

- Describe how this human-understandable rules about facial expressions perform. Are the resulting rules consistent to what you would expect? Do these rules make sense?

5. **Creative Challenge (25%)**
    - Improve FER2013 performance using at least 2 or more of:
        - Hyperparameter tuning
        - Feature engineering
        - Feature selection
        - Class balancing
        - Data augmentation
        - Data curation and cleaning
        - Label verification
        - Use of KDEF-derived features or insights
            - Use KDEF to identify informative features.
            - Learn geometric features on KDEF.
            - Emotion prototypes from KDEF.
            - Domain-invariant features
            - Determine which features transfer best.
            - Data cleaning
        - Any other justified approach

6. **Run your models in a real-time a real-time emotion recognition program with your laptop (10%)**
    - Adapt your best model for real-time processing with your camera webcam (basic demo file provided)
    - Report:
        - Did your models perform well for this applications? Yes/no explain why
        - What problems did you run into?
        - If you and what fine tuning did you need to do to improve your models for this task.

Cross-generalization experiments

To ensure a fair comparison between the FER→KDEF and KDEF→FER settings, all images must be converted to the FER image format prior to training and evaluation.

KDEF preprocessing

Apply the following preprocessing steps to all KDEF images:

1. Detect and crop the face region.

2. Convert the image from RGB to grayscale.
3. Resize the image to the FER resolution.
4. Apply the same normalization procedure used for FER.

Rationale

FER images are natively grayscale and low resolution, whereas KDEF images are RGB and substantially higher resolution. Converting KDEF images to the FER format reduces domain differences related to image acquisition, colour representation, and resolution.

The reverse operation is not recommended, since it is not possible to recover the missing colour and high-frequency information from FER images. Therefore, a common FER-based representation will be used for both FER→KDEF and KDEF→FER experiments.

Experimental Protocol

- **FER→KDEF:** Train on FER and test on pre-processed KDEF images.
- **KDEF→ same FER:** Train on pre-processed KDEF images and test on FER images.
- Use the image size, grayscale representation, and normalization pipeline in both experiments.

The objective is to evaluate cross-dataset generalization while minimizing confounding factors introduced by differences in image format and preprocessing. Face cropping can be performed using one of OpenCV's face detectors. The detected face bounding box should be expanded slightly in each direction before cropping to ensure that the complete facial region, including the forehead and jawline, is preserved. The same cropping procedure must be applied to all KDEF images prior to grayscale conversion and resizing. A basic preprocessing algorithm is included in the assignment for reproducibility and can be improved by the student it considered necessary for the creative challenge if properly justified.

Report

For reporting we expect a report of maximum 8 pages. The report should clearly describe

- Main results for each of your experiments
- Discussion comparing which features and models worked best/did not work and why
- Insights on the dataset: What are specific challenges for each dataset? Are these challenges present in both of the datasets or only on one of them?
- Discussion on geometric features on fuzzy model.
- Descriptions on the features and processing performed on the creative challenge.
- Challenges and adaptation of models for the real-time demo.

Add an appendix with confusion matrices for each of the reported models assignment. You can refer to them in your report. In the appendix avoid long explanations.

What to deliver?

You have to deliver a .zip file with the following files:

1. The report in PDF.
2. Jupyter notebook(s) where you trained your models and do the performance measurements.
3. A set of .pkl files which contains your trained models.
4. And a demo python file which we can call from the command line to execute the code and which will prompt the camera feed where the teaching team can test your best performing algorithm.

If the feature extraction file is too big you optionally can put the feature extraction in a separate python file. This is the only exception, the rest of the code should be implemented in the notebook.

What are we grading?

The assignment will be graded on the following aspects:

1. **Understanding and justification of design choices**
    - Are the selected features, models, and preprocessing steps properly motivated?
    - Are the design decisions supported by experimental evidence?
2. **Understanding of the datasets**
    - What are the main characteristics of FER2013 and KDEF?
    - What challenges are specific to each dataset?
    - Which challenges are shared by both datasets?
3. **Feature analysis**
    - Which feature representations perform best?
    - Which features are most robust across datasets?
    - Can the observed behavior be explained?
4. **Model analysis**
    - Which classifiers perform best and why?
    - What are the strengths and limitations of each model?
    - For which types of applications might each model be appropriate?
5. **Evaluation and interpretation of results**
    - Are the results analyzed critically?
    - Are metrics, confusion matrices, and examples used appropriately?
    - Are limitations and failure cases discussed?
6. **Cross-dataset generalization**
    - Is domain shift correctly identified and discussed?
    - Are the differences between FER2013 and KDEF properly analyzed?
    - Are transfer results interpreted meaningfully?
7. **Implementation quality**
    - Is the code well organized and readable?
    - Is the implementation reproducible?
    - Are notebooks and scripts clearly documented?

For every component of the assignment, the grading will be performed as follows

Component	Weight
FER2013 baseline	20%
KDEF study	15%
Cross-dataset evaluation	20%
Fuzzy classifier	10%
Creative challenge	25%
Real time demo	10%

Code

The report and code should be packed as a single zip file. The packages for this algorithm should be the standard ones used through the course (numpy, scikit learn, scipy, open CV. etc.).

Python libraries

For this assignment we will use the same libraries as usual. New libraries that can be used are Dlib and/or Open CV for handling the camera and landmark detection (if applicable). For the fuzzy classifier PyFUME (https://github.com/CaroFuchs/pyFUME), which needs Numpy (Numpy 1.24.4). HIGHLY recommended new environment for this. Otherwise you may break the environment you use for the other models. An additional class thagt can be used is imbalanced learn. For the fuzzy model you also can use the ANFIS library. Beware that this may need some extra work to make it work for classification fuzzy systems. More information on this later.

Use of AI

As discussed in the first lecture, it is not forbidden the use of AI as long as you are honest and report how and for which parts of your work have been used. Despite of this, we want to evaluate your knowledge and learnings from the assignment, so avoid using the AI model to generate everything as this will be considered when grading. Remember to use the default technology statement, using the text below.

Replace the text in capital letters with the requested information.

“During the preparation of this work, I/We used [NAME TOOL / SERVICE / VERSION OF AI TOOL] in order to [REASON]. The following parts of the assignment were affected/generated by AI tool usage: [INTRODUCTION / METHODS / xxx, DISCUSSION]. After using this tool/service, [NAME STUDENT(S)] evaluated the validity of the tool’s outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [NAME STUDENT (S)] take(s) full responsibility for the content of their work.

References

[R1] Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., ... & Bengio, Y. (2013, November). Challenges in representation learning: A report on three machine learning contests. In *International conference on neural information processing* (pp. 117-124). Berlin, Heidelberg: Springer Berlin Heidelberg.

[R2] Goeleven, E., De Raedt, R., Leyman, L., & Verschuere, B. (2008). The Karolinska directed emotional faces: a validation study. *Cognition and emotion*, *22*(6), 1094-1118.